

OMAR IBRAHIM

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Education

Carleton University

Sep. 2024 – Apr. 2028

Bachelor of Engineering in Software Engineering

Ottawa, Ontario

- **Relevant Coursework:** Data Structures, Digital Systems, Computer Architecture, Software Engineering

Technical Skills

Security: Penetration Testing, Web App Security, Burp Suite, Kali Linux, Postman, TLS, PKI, ML-KEM, ML-DSA

Infrastructure: Linux, Docker, Nginx, Cloudflare, Tailscale, UFW, GitHub Actions CI/CD, PostgreSQL, Redis

Languages: Python, Bash, TypeScript, SQL, C++, Java

AI/ML: Agentic AI, LLM Integration, RAG, Ollama, Computer Vision, YOLOv8, OpenCV

Security Experience

Security Analyst & Software Developer

Oct. 2025 – Jan. 2026

Journale AI (Competitech Inc.)

Ottawa, Ontario

- Pentested an **AI dialogue platform** and **game-engine SDKs**, finding **6 critical flaws** in backend **REST APIs**.
- Found and fixed a **Stripe API** bypass that granted paid subscriptions for free, **patched in under 5 hours**.
- Proved impact with **Burp Suite**, **Postman**, and **cURL**, and drove fixes under **responsible disclosure**.

Penetration Tester

Mar. 2026 – Apr. 2026

cuHacking, BearHacks, CUMSA

Ottawa, Ontario

- Security-tested **3 production web platforms** before launch, incl. **Ottawa's largest hackathon**, cuHacking.
- Audited **cumsa.ca** and rewrote its **authentication logic**; served **on call** during live hackathon weekends.

Technical Workshop Lead

March 2026

Institute of Electrical and Electronics Engineers (IEEE)

Ottawa, Ontario

- Delivered **4 technical workshops** to **100+ students** across campus for a 2-week workshop series at uOttawa.
- Taught **Linux** fundamentals, **Docker**, **Github Actions** CI/CD, **AI/Machine Learning**, and **Cybersecurity**.
- Built a **lab environment** with isolated per-student **Docker containers** and **custom challenges** to test students.

CTFs & Awards

- **1st Place** – uOttawa Cyber Club CTF: **top individual score** in web, crypto, and reversing Apr. 2026
- **Most Flags Found** – BSides Ottawa CTF: ran **parallel AI sub-agents** to lead at **~4,000 points** Apr. 2026
- **1st Place** – IEEE uOttawa × Hack The Box CTF: team took **64 of 67 flags**, **2nd individually** Feb. 2026
- **IEEE Engineer of the Year**; **4 hackathon awards** incl. Canada's largest AI hackathon (**1,000+ hackers**) 2026

Projects

🔗 **Post-Quantum TLS Scanner** | *Python, TLS 1.3, ML-KEM, ML-DSA, Nginx, Docker*

Oct. 2026

- Hand-built **TLS 1.3 ClientHello** probes detecting **ML-KEM hybrid key exchange**, independent of OpenSSL.
- Graded **49 sites**; only **10 of 23** critical Canadian services negotiate **post-quantum key exchange**.
- Built a hardened **Nginx lab** with an **ML-DSA-65** cert chain; **31 tests** in CI, validated against **OpenSSL**.

Hardened Linux Infrastructure | *Ubuntu, Docker, Nginx, Cloudflare, Tailscale, UFW*

Dec. 2025 – Present

- Run an **Ubuntu VPS** hosting **5+ sites** over HTTPS for **8,000 unique visitors** and **300k requests monthly**.
- Hardened servers with **default-deny firewalls**, **key-only SSH**, **Tailscale**, and **auto security updates**.
- Installed Ubuntu on a laptop with a **dead screen fully over SSH**, making it a **24/7 headless security lab**.

🔗 **QueryForge – Private NL-to-SQL Agent** | *Python, DuckDB, Ollama, SQL*

Mar. 2026

- Built an AI agent that answers plain-English questions over a database, hitting **98% accuracy** on **172 test cases**.
- Ran fully **offline on CPU** through **Ollama**, so sensitive data **never leaves the machine** and API cost is **zero**.